AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MST129
Applied Calculus

Midterm Examination (MTA)
Spring Semester 2024/2025

Date: 7/ April /2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed: 90 minutes**

Instructions:

- Total Marks: **60**
- This exam consists of **two** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Scientific Calculators** are allowed.

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Q–1: [5×3 marks] choose the correct answer:

1.1: The equation of the line parallel to the line y= -2x and passing through (3, 5) is:

a) y= -2x+11
b) $y = -2x + 1$
c) y= -1/2 x+13/2
d) y= 1/3 x +7/2
e) None of the above

1.2: The slope of the graph of $f(x) = (4x - 10)^5 - (3x - 1)^2$ at x=3 is:

a) 48
b) 272
c) 320
d) 368
e) None of the above

1.3: The second derivative of $f(x) = x^7 + x^3$ at $x = 1$ is:

a) 84
b) 10
c) 60
d) 48
e) None of the above

1.4 : The critical number of the function $f(x) = (4 - x)^5$ is:

a) 4
b) -4
c) $0, 4$
d) $0, -4$
e) None of the above

1.5: The average rate of change of f(t)= $t^2 + 3t - 7, when\ t = 5$ is:

a) 5 km/hr^2
b) 25 km/hr^2
c) 33 km/hr^2
d) 13 km/hr^2
e) None of the above

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Each question is worth 15 marks. Answer the following questions:

Q–2a) Let $f(x) = -x^2 + x - 1$ and $g(x) = x - 5$, h(x)= $\sqrt{x-7}$

i) [4 marks] Find the domains of $f(x)$ and $h(x)$ in interval notation.

ii) [5 marks] Find$(g \circ f)(x)$.

b) [6 marks Find the points of intersection of the curves f(x) and g(x)

Q–3:

a) [7 marks] Find the equation of the tangent line to the curve f(x)= $3x^2 - 2x + 1$ at x=1

b) [8 marks] Suppose that t seconds after starting to run, an individual's pulse rate (in beats/min) is given by p(t)= 56 +$2t^2$ -t, where $0 \leq t \leq 7$. Find the rate of change of p(t) with respect to t at t= 2.

Q–4: Let $f(x) = x^3 - 27x$

a) [4 marks] Find $f'(x)$ and$f''(x)$.

b) [11 marks] Find the intervals on which f is increasing or decreasing and find the local maximum and minimum values of $f(x)$, if exist, by using the first derivative test.

End of questions

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